

AKSHAY MERUGU

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SUMMARY

Software engineer with an M.S. in Computer Science, experienced in Python, C++, Linux, backend development, and production authentication systems. Builds secure services and APIs while applying machine learning, retrieval-augmented generation, and data-focused tooling, with exposure to Docker and CI/CD workflows.

EDUCATION

University of Alabama at Birmingham

Master of Science, Computer Science

• **GPA:** 3.9/4.0

• **Coursework:** Machine Learning, Advanced Algorithms, Software Engineering, Data Mining, Database Systems

August 2024 - May 2026

CMR Institute of Technology

Bachelor of Technology, Computer Science

• **GPA:** 3.3/4.0

August 2020 - June 2024

EXPERIENCE

Holiday Channel

July 2026 - Present

Full-Stack Developer Intern

- Designed and implemented a multi-tenant authentication system using NestJS and Auth0 Organizations, supporting platform-admin and per-seller login flows from a single OAuth application through the OIDC authorization-code flow with CSRF-safe state and nonce handling.
- Identified and flagged critical security gaps across the platform, including unauthenticated production API endpoints and an exposed database credential in a separate repository, prioritizing them for remediation.
- Established a new NestJS service from scratch by managing scaffolding, dependencies, and environment configuration, creating the foundation for the company's Backend-for-Frontend (BFF) layer.
- Audited and mapped a 6-repo e-commerce marketplace platform (Next.js storefront, Next.js API backend, seller dashboard, Supabase/Postgres migrations) to document data flow, authentication gaps, and cross-service dependencies.

Suraksha Child Development Center

January 2024 - January 2025

Full-Stack Developer (Freelance)

- Architected, deployed, and maintained a production web platform using Python with Flask, MySQL, and JavaScript, including the relational schema and backend APIs supporting user management and operational workflows.
- Designed secure authentication and role-based access workflows using bcrypt password hashing, session management, and administrator controls.
- Built appointment scheduling and service management modules with automated email notifications and booking confirmations, working directly with stakeholders to turn requirements into shipped features.

PROJECTS

Attention-Driven Crypto Volatility Forecasting

March 2026

- Built an end-to-end Python machine learning pipeline that forecasts 7-day cryptocurrency volatility and classifies market risk regimes from combined market and public-attention signals.
- Engineered predictive features from Yahoo Finance market data and the Wikipedia Pageviews API, including rolling volatility, trend indicators, and attention metrics.
- Containerized the forecasting workflow with Docker and deployed an interactive Streamlit interface for real-time analysis and model output visualization.

DocTalk

January 2026

- Designed and implemented a retrieval-augmented generation (RAG) system using large language models for conversational question-answering over PDF documents.
- Developed document ingestion and chunking pipelines with semantic vector indexing using ChromaDB and Gemini embeddings, supporting reliable retrieval for downstream generation.
- Integrated Ollama-hosted LLaMA 3.2 models for fully local inference and privacy-preserving responses, tuning retrieval through semantic search and context-aware prompting.

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, TypeScript, Kotlin, SQL, HTML5, CSS3

Backend & Web: NestJS, Node.js, Next.js, Flask, FastAPI, Django, REST APIs, Vue.js, Streamlit

Auth & Security: Auth0, OAuth 2.0 / OIDC, role-based access control, session management, bcrypt, CSRF protection

AI & Machine Learning: Scikit-learn, Pandas, NumPy, LLMs, RAG, ChromaDB, Gemini API, Ollama, LLaMA, Deep Learning, TensorFlow, PyTorch, Reinforcement Learning, Generative AI, MLOps

Databases: PostgreSQL, Supabase, MySQL, MongoDB, Redis, Amazon DynamoDB, Amazon RDS

Cloud & DevOps: Docker, Git, GitHub, CI/CD, Linux, Windows, Unix Shell, Azure, Kafka, Power BI